Real-Time Spectral Photoelasticity via Stochastic Hero-Wavelength Ray Marching and Stress-Aware Temporal Reprojection

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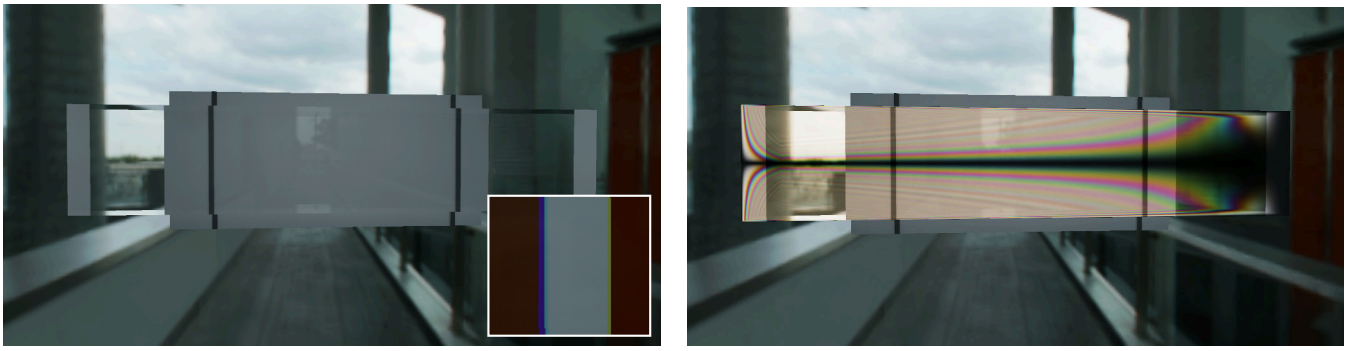


Figure 1: *An acrylic beam rendered with our system at* 1920×1080 *on an NVIDIA RTX 3070 Laptop GPU at* >120 *FPS. A diffuse grey backdrop with vertical reference strips sits behind the geometry, isolating the high-frequency chromatic dispersion and the internal isochromatic fringes against the path-traced environment.* ***Left:*** *no applied load.* ***Inset:*** *magnified detail of the right-end refractive boundary under a grazing view, revealing the Cauchy chromatic dispersion as blue and yellow edge fringes (not visible at normal incidence).* ***Right:*** *the same beam under an interactively applied point load; isochromatic fringes appear throughout the volume.*

Abstract

*We address rendering photoelastic stress fringes at real-time rates. The state-of-the-art method, Bußler et al.'s GPU ray-caster [BES15], integrates the Jones Calculus along each ray for 16 fixed wavelengths, giving frame times of 231 to 6 400 ms and no interactive stress updates. Adapting Hero Wavelength Sampling [WND*14] to this integral with four SIMD-vectorised wavelengths rather than a deterministic 16-step loop yields an unbiased estimator with a theoretical* $4\times$ *wavelength reduction. Interactivity introduces a critical temporal challenge: fringe patterns shift instantaneously under dynamic loads while geometry remains static. This violates the geometric motion assumptions of standard temporal anti-aliasing (TAA), causing severe persistent ghosting. We resolve it with a stress-aware history rejection gate driven by the temporal gradient of the local principal stress difference. To isolate our algorithmic gains from simulation constraints, we evaluate on two interactive analytical stress fields and one pre-computed FEM tensor volume (the Kirsch plate), matching prior offline methods' data setup. The framework achieves over 120 FPS at 1080p on consumer hardware, an* $8.55\times$ *algorithmic speedup over the deterministic baseline on identical hardware, and over* $100\times$ *absolute improvement against the prior published state of the art.*

CCS Concepts

• ***Computing methodologies*** → ***Ray tracing;*** *Antialiasing;* • ***Human-centered computing*** → *Scientific visualization;*

1. Introduction

Photoelasticity is the phenomenon by which mechanical stress temporarily renders a transparent material birefringent. A stressed poly(methyl methacrylate) (PMMA) sample between crossed polarising filters displays coloured interference fringes whose distribution encodes the principal stress difference at every point inside the material [DR78]. For engineers this is not primarily a visual ef-

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fect but a measurement technique: the fringe count at a location is a physical quantity.

Rendering these fringes requires integrating the Jones Calculus along every camera ray through the stressed volume, accumulating the wavelength-dependent phase retardation at each step. Bußler et al. [BES15] implemented this faithfully, with Fresnel refraction and reflection at the boundary, validated against experimental polariscope photographs. The method works but is computationally bound by design: producing chromatic fringes requires the full Jones integral at each of 16 deterministically sampled wavelengths per ray step, yielding 231 to 6 400 ms on a GTX 760 depending on dataset size. True interactivity needs a fundamentally different sampling approach, which would open the technique to real-time engineering analysis and VR training.

The key focus here is not peak frame rate but frame budget. For a 90 Hz VR display with an 11.1 ms budget, the deterministic spectral loop alone exceeds the entire frame allowance on current consumer hardware (Section 8), leaving no room for the rest of the engine. Interactivity also does not diminish quantitative utility: the stress tensor grid that powers the optics remains stored on the GPU for numerical readback, and the rendered fringe orders correspond closely to analytic values, within 0.4% at the neutral axis (Table 3). What changes with interactivity is how the instrument is used; an offline render for each configuration becomes a live load sweep.

Bußler's 16-wavelength loop, a deterministic spectral quadrature for the tri-stimulus colour response, can be replaced by Monte Carlo quadrature with temporal averaging, for which we use Hero Wavelength Sampling [WND*14], propagating only four wavelengths per frame instead of 16. The adaptation is non-trivial because the "spectral integrand" is a complex polarisation state rather than surface reflectance, though the mathematical structure is directly analogous. The result is a `float4`-vectorised Jones integration that takes a quarter of the instruction cycles and converges under temporal accumulation.

Interactivity exposes a secondary temporal challenge. When the user changes the applied load, the internal stress field and hence the fringe pattern change everywhere inside the volume, yet the boundary geometry remains static. Standard temporal anti-aliasing (TAA) cannot detect this internal optical shift because geometric motion vectors are zero, so it blends the new fringes with the stale history buffer, producing a smeared, unreadable image. The fix is to track the per-pixel local stress magnitude between frames and reject history when its temporal gradient exceeds a dynamically derived threshold.

Summary of contributions.

- We apply Hero Wavelength Sampling to the Jones Calculus volume integral (Section 5.2), giving the first stochastic estimator for spectral photoelastic rendering: it is unbiased, SIMD-friendly, and requires no change to the downstream path tracer.
- We identify why standard TAA fails silently on interactive, optically dynamic volumes, and show that a stress-gradient gate in place of motion vectors resolves the resulting ghosting (Section 6).
- We introduce a pipeline visualising both interactive analytical stress fields and complex FEM tensor volumes. With a validation strategy matching the FEM baseline of prior offline methods, we demonstrate a $> 100\times$ absolute speedup against the published GTX 760 timing, combining algorithmic improvement with a decade of hardware progress (Section 8).

To facilitate reproduction, the complete source code, scene files, FEM tensor data, environment maps, and the capture and metric scripts behind every figure and table in this paper are included in the supplementary material, and will be released in a public repository [Cod].

2. Related Work

2.1. Photoelastic rendering

Aben [Abe79] established the framework we work within: a Jones vector $\mathbf{E}$ propagates through a continuously stressed medium by integrating $d\mathbf{E}/dz = \mathbf{G}\mathbf{E}$, with $\mathbf{G}$ set by the local stress tensor. Bußler et al. [BES15] implemented this in CUDA with Runge–Kutta integration, added Fresnel refraction and reflection, and produced the first images resembling real polariscope photographs. Theirs is our direct baseline: we keep their core optics model, replace the deterministic 16-wavelength loop with a stochastic estimator, and add the interactive physical coupling their pre-computed static 3D textures preclude.

On the inverse side, NeST [D*25] reconstructs internal stress fields from multi-axis polariscope measurements with a differentiable Jones Calculus forward model, at the cost of over 400 object rotations, 6 400 captured images per object, and more than two hours on an NVIDIA A100. That forward model is monochromatic and static, so it cannot handle changing loads or render chromatic fringes. The inverse and forward problems compose: NeST could supply a reconstructed field that our renderer visualises interactively, completing a pipeline neither achieves alone.

2.2. Birefringent volume rendering

Weidlich and Wilkie [WW08] derived the Fresnel coefficients and ray-splitting formulas for uniaxial crystals with constant optical properties; Latorre et al. [LSG12] extended this to biaxial media. Both treat crystalline materials with spatially uniform birefringence; our stress-varying case requires integrating the Jones matrix continuously along the ray. Steinberg [Ste20] presented a closed-form analytic expression for light transport in inhomogeneous anisotropic media (e.g. liquid crystals and stressed elastic fluids), posing the problem as a differential equation. We instead discretise the Jones integral over a GPU-resident stress grid, which admits real-time interactive updates.

2.3. Spectral sampling in rendering

The spectral integral in path tracing was long treated deterministically, evaluating reflectance and emission at a fixed set of wavelengths per ray. Wilkie et al. [WND*14] introduced Hero Wavelength Sampling, where one "hero" wavelength is drawn randomly and companions sit at equidistant intervals covering the visible spectrum; propagating this four-wavelength packet per ray gives

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an unbiased estimator that converges under frame averaging; the technique is now standard in offline renderers such as PBRT-v4 [PJH23]. Our key observation is that the spectral dimension of photoelastic rendering has the same structure: a smooth integrand over wavelength, accumulated per pixel over time.

2.4. Temporal reprojection

Standard TAA [YLS20] accumulates frames assuming radiance at a world-space point is constant between them, discarding history when motion vectors indicate the scene moved. SVGF [SKW*17] adds spatiotemporal variance-guided filtering for path-traced illumination but relies on the same brightness-constancy assumption. Iglesias-Guitian et al. [IGMM22] addressed temporal denoising for volumetric path tracing without reliable G-buffers, yet still assume stable volume properties between frames. Our setting violates that assumption: the volume's optical properties change every frame during interaction. None of these methods handle this case, necessitating our stress-gradient rejection criterion.

3. Background

3.1. Stress-optic law

Inside a transparent stressed material the in-plane stress tensor acts as a local birefringent retarder [DR78, Abe79]. Under plane stress conditions [DR78], which model thin plates, the in-plane tensor components (normal σ_{xx}, σ_{yy}, shear τ_{xy}) are:

$$\boldsymbol{\sigma}' = \begin{pmatrix} \sigma_{xx} & \tau_{xy} \\ \tau_{xy} & \sigma_{yy} \end{pmatrix} \tag{1}$$

the principal axis angle is $\alpha = \frac{1}{2}\operatorname{atan2}\big(2\tau_{xy}, \sigma_{xx} - \sigma_{yy}\big)$ and the phase retardation over a step Δs is

$$\delta(\lambda) = \frac{2\pi C}{\lambda}\,\Delta\sigma\,\Delta s, \qquad \Delta\sigma = \sqrt{(\sigma_{xx} - \sigma_{yy})^2 + 4\tau_{xy}^2} \tag{2}$$

where C is the material stress-optic coefficient.

3.2. Jones matrix integration

Each retardation step contributes a Jones matrix

$$\mathbf{J}(\delta, \alpha) = \begin{pmatrix} c_\delta + i s_\delta c_{2\alpha} & i s_\delta s_{2\alpha} \\ i s_\delta s_{2\alpha} & c_\delta - i s_\delta c_{2\alpha} \end{pmatrix} \tag{3}$$

where $c_\delta = \cos(\delta/2)$, $s_\delta = \sin(\delta/2)$, $c_{2\alpha} = \cos(2\alpha)$, $s_{2\alpha} = \sin(2\alpha)$. The exit field after N steps is the path-ordered product $\mathbf{E}_{\text{out}} = \prod_{i=N}^{1} \mathbf{J}_i\,\mathbf{E}_{\text{in}}$, and the crossed analyser extracts

$$I(\lambda) = \left| -\sin\phi\, E_x + \cos\phi\, E_y \right|^2 \tag{4}$$

where ϕ is the polariser angle and the crossed analyser transmits the component along $\phi + 90^\circ$.

4. Stress Field Evaluation

The optical integration requires a volumetric stress field in a `Texture3D`. We evaluate exact closed-form fields on the GPU and support high-resolution tensor grids baked from offline Finite Element Method (FEM) simulations, so the pipeline can prototype temporal changes interactively while still validating complex engineering geometries.

4.1. Cantilever beam

Following Timoshenko and Goodier [TG51], the exact stress distribution for a beam of length L, height H, depth B and second moment of area $I = BH^3/12$, loaded by a transverse point force P at x_{load}, with y the vertical distance from the neutral axis, is:

$$\sigma_{xx} = \frac{P\,(x_{\text{load}} - x)\,y}{I}, \quad x < x_{\text{load}} \tag{5}$$

$$\tau_{xy} = \frac{P\,Q}{I\,B}, \quad Q = \frac{B}{2}\left(\frac{H^2}{4} - y^2\right), \quad x < x_{\text{load}} \tag{6}$$

with $\sigma_{yy} = 0$ (plane stress) and all components zero for $x \geq x_{\text{load}}$ or $|y| > H/2$. The load position x_{load} maps to the cursor each frame via the host's `onMouseEvent`, making the field interactive at sub-frame cost.

4.2. Diametral disk compression

The closed-form Flamant solution for a disk of diameter D and thickness B under a diametral load P applied at the top and bottom poles [Fro48], with $R = D/2$, $r_1^2 = x^2 + (R-y)^2$, and $r_2^2 = x^2 + (R+y)^2$, is given by:

$$\sigma_{xx} = \frac{2P}{\pi B}\left[\frac{1}{D} - \frac{x^2(R-y)}{r_1^4} - \frac{x^2(R+y)}{r_2^4}\right] \tag{7}$$

$$\sigma_{yy} = \frac{2P}{\pi B}\left[\frac{1}{D} - \frac{(R-y)^3}{r_1^4} - \frac{(R+y)^3}{r_2^4}\right] \tag{8}$$

$$\tau_{xy} = \frac{2P}{\pi B}\left[\frac{x(R-y)^2}{r_1^4} - \frac{x(R+y)^2}{r_2^4}\right] \tag{9}$$

Singularities at the load poles are regularised by clamping r_1^2 and r_2^2 to a minimum of $\varepsilon = 10^{-6}$.

The choice of grid resolution is an accuracy–memory trade-off, evaluated empirically in Section 8.5.

5. Stochastic Spectral Photoelasticity

5.1. Why the deterministic loop is the bottleneck

Getting colour out of Equation 4 requires enough wavelengths to reconstruct the visible spectrum. Bußler et al. use 16 uniform samples in $[380, 700]$ nm, converting to RGB via the CIE 1931 colour-matching functions. At 1920×1080 and $N = 32$ steps this is roughly 10^9 complex matrix-vector products per frame, far beyond what consumer hardware can execute within the 11.1 ms VR budget. Our estimator reduces that 16-step multiplier to a single vectorised stochastic evaluation of four wavelengths at once, while remaining unbiased.

5.2. Hero Wavelength Stratified Estimator

We partition the visible spectrum into $M = 4$ strata of 100 nm, spanning a broader range, $[380, 780]$ nm, than the deterministic baseline so deep red isochromatic fringes are captured, and draw one sample per stratum every frame [WND*14]. For pixel $\mathbf{x}$ at frame f the hero wavelength comes from a per-pixel hash with a golden-ratio temporal offset:

$$\lambda_h = 380 + 400 \cdot \operatorname{frac}\big(h(\mathbf{x}) + g \cdot f\big) \tag{10}$$

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where $h(\mathbf{x}) \in [0,1)$ is a standard procedural hash and g the golden-ratio conjugate, giving each pixel a low-discrepancy temporal wavelength sequence; the constants are given in Section 7. Three companions complete the strata at fixed spectral offsets:

$$\lambda_k = 380 + \big((\lambda_h - 380) + k \cdot 100\big) \bmod 400, \quad k \in \{1,2,3\} \tag{11}$$

Every frame therefore has one sample in each 100 nm band. The four wavelengths are packed into a `float4` and the retardation (Eq. 2) vectorised:

$$\boldsymbol{\delta} = \frac{2\pi C \,\Delta\sigma\, \Delta s \cdot \lambda_{\text{ref}}}{\boldsymbol{\lambda}}, \quad \lambda_{\text{ref}} = 580.0\,\text{nm} \tag{12}$$

where λ_{ref} is a normalisation constant chosen for implementation convenience: the coefficient supplied to the implementation equals C/λ_{ref}, so a coefficient normalised at the sodium D-line (589.3 nm) is multiplied by $\lambda_D/\lambda_{\text{ref}}$ before use. All four Jones evaluations execute as one `float4` SIMD sequence, giving

$$\mathbf{c}_{\text{fringe}} = \tfrac{1}{4} \sum_{k=0}^{3} \mathbf{r}(\lambda_k)\, I(\lambda_k) \tag{13}$$

where $\mathbf{r}(\lambda)$ is a piecewise-linear CIE 1931 approximation.

5.3. Variance and convergence

Let $f(\lambda) = \mathbf{r}(\lambda) I(\lambda)$ denote the integrand. Write μ_k, σ_k^2 for the mean and variance of f within stratum k, and μ for the global spectral mean.

By the standard stratified sampling result [PJH23]:

$$\begin{aligned} \text{Var}[\hat{I}_{\text{HW}}] &= \frac{1}{16} \sum_k \sigma_k^2 \\ &\le \frac{1}{16} \left(\sum_k \sigma_k^2 + \sum_k (\mu_k - \mu)^2 \right) = \text{Var}[\hat{I}_{\text{MC}}] \end{aligned} \tag{14}$$

where $\hat{I}_{\text{MC}}$ is the independent four-sample estimator. The variance reduction equals the between-stratum variance, which grows with the number of visible fringe orders. Optical phase retardation is accumulated *spatially*, inside each frame's path-ordered Jones product (Section 3.2), where per-wavelength phase is fully tracked. Across wavelengths and frames we deliberately average post-modulus intensities: broadband illumination is temporally incoherent, distinct wavelengths carry no mutual phase relation, and the measurable quantity is the spectral power integral, for which $\hat{I}_{\text{HW}}$ is unbiased. Under temporal accumulation the estimator converges at the standard $O(1/\sqrt{T})$ Monte Carlo rate, where T is the number of accumulated frames, with stratification reducing the variance constant.

5.4. Cauchy dispersion at boundaries

At the exit surface, the IOR follows the Cauchy equation [BW99, Ste19]:

$$n(\lambda) = n_0 + \frac{C_0}{\lambda^2}, \quad \lambda \text{ in nm} \tag{15}$$

with empirical coefficients $n_0 = 1.4933$ and $C_0 = 3942\,\text{nm}^2$ for PMMA. This dependence ($\Delta n \approx 0.021$ across the visible spectrum) produces chromatic aberration. The stochastic hero wavelength is not used here: it creates firefly noise along silhouettes that does not average cleanly. Dispersion is instead computed once per pixel at three fixed wavelengths $\lambda_R = 650$, $\lambda_G = 550$, $\lambda_B = 450$ nm, three rather than four so that they map one-to-one onto the environment map's RGB channels. The angular deviation from the 550 nm baseline is

$$\Delta\theta(\lambda) = \arcsin\big(n(\lambda)\sin\theta_i\big) - \arcsin\big(n_{550}\sin\theta_i\big) \tag{16}$$

where θ_i is the exit angle from the ray-AABB slab method, with the exit normal recovered analytically as the axis with the smallest per-axis far-intersection t. The environment map is sampled three times with per-channel UV offsets proportional to $\Delta\theta$, and the dispersed background $\mathbf{c}_{\text{disp}}$ blended in proportion to $\sin\theta_i$:

$$\begin{aligned} \mathbf{c}_{\text{out}} = \text{lerp}(&\mathbf{c}_{\text{bg}}, \mathbf{c}_{\text{disp}}, \sin\theta_i) \\ &\cdot (12\,\mathbf{c}_{\text{fringe}} + 0.1) \end{aligned} \tag{17}$$

The scale 12 and bias 0.1 are empirical constants calibrated against luminance in physical polariscope photographs; the shader applies the equivalent scale 3 to the unnormalised stratum sum of Eq. 13. The weight $\sin\theta_i$ vanishes at normal incidence, where a flat slab produces no lateral chromatic shift, so dispersion appears only at oblique angles. Because the Cauchy coefficients are material constants, this boundary fringing is stress-independent: it stays fixed during load interaction while the internal isochromatic fringes change.

6. Stress-Aware Temporal Reprojection

6.1. The TAA failure mode

The hero wavelength estimator is noisy per frame and requires temporal accumulation to converge. In the static case (load held constant, no interaction) this is straightforward; the problem arises when the user changes the load.

When the load point moves, every pixel whose ray crosses the beam volume evaluates a different fringe colour next frame. The geometry has not moved and motion vectors are zero, so standard TAA holds its full history ($w = 0.95$), blending a stale fringe pattern with the new one. Figure 3 shows the resulting degradation in structural similarity (SSIM) during dynamic events. While theoretical decay dictates a long tail, we observe severe ghosting above the ambient noise floor for approximately **25 to 28 frames** (∼450 ms at 60 FPS) after a load stabilises, long enough to make the interactive mode unusable.

Neighbourhood colour clamping does not help: the new fringe colours are not outliers relative to the neighbourhood, merely different from last frame in a way invisible to any purely geometric reprojection criterion.

6.2. Stress-Gradient History Gate

Our solution tracks the stress state at each pixel's surface hit point across frames, using the Euclidean norm of the unique in-plane components:

$$\Sigma_t = \sqrt{\sigma_{xx}^2 + \sigma_{yy}^2 + \tau_{xy}^2} \tag{18}$$

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Algorithm 1 Stress-Aware Temporal Reprojection

Require: pixel $\mathbf{p}$, current colour $\mathbf{c}_t$, history buffers $\mathbf{H}_{t-1}$, $\mathbf{\Sigma}_{t-1}$
$\Sigma_t \leftarrow \sqrt{\sigma_{xx}^2+\sigma_{yy}^2+\tau_{xy}^2}$ (at surface hit point of $\mathbf{p}$)
$\Delta \leftarrow |\Sigma_t - \mathbf{\Sigma}_{t-1}[\mathbf{p}]|$
$\tau \leftarrow 0.005\,\Sigma_t + 0.1$
if $\Delta > \tau$ **then**
  $w \leftarrow 0$
else
  $w \leftarrow 0.95$
end if
$\mathbf{c}_{\text{out}} \leftarrow (1-w)\,\mathbf{c}_t + w\,\mathbf{H}_{t-1}[\mathbf{p}]$
$\mathbf{H}_t[\mathbf{p}] \leftarrow \mathbf{c}_{\text{out}}$; $\mathbf{\Sigma}_t[\mathbf{p}] \leftarrow \Sigma_t$

stored in an `R32F` texture (8 MB at 1080p). The blend weight is gated on its temporal change:

$$w = \begin{cases} 0 & \text{if } |\Sigma_t - \Sigma_{t-1}| > \tau_t \\ \alpha_{\text{base}} & \text{otherwise} \end{cases} \qquad (19)$$
$$\tau_t = \gamma\Sigma_t + \beta$$

with $\alpha_{\text{base}} = 0.95$, $\gamma = 0.005$, $\beta = 0.1$. The threshold τ_t adapts to local stress magnitude, avoiding false rejections near load singularities where small positional changes give large gradients but negligible optical effect. Algorithm 1 details the per-pixel procedure.

Because the gate keys on per-pixel stress change rather than motion vectors, it is not specific to load changes: rigid specimen motion such as stage rotation changes Σ_t exactly where fringes move, triggering the same rejection. For in-plane rotation by θ, rays transform rigidly into the specimen frame and the fast-axis angle becomes $\alpha + \theta$; Section 8.4 evaluates this jointly with camera and load dynamics.

7. Implementation

Our `PhotoelasticityPass` is a GPU compute pipeline in the NVIDIA Falcor [B*20] render graph (Figure 2), written in Slang. Each frame runs three sub-passes in sequence. The physics pass is dispatched over the voxel grid (32^3 analytical, 128^3 FEM); its threads evaluate the closed-form equations of Section 4 or fetch pre-baked FEM tensors, writing $(\sigma_{xx}, \sigma_{yy}, \tau_{xy})$ to an `RWTexture3D<float4>`. The optics pass is dispatched over the framebuffer: pixels hitting the specimen march the N-step Jones integration of Section 5 (default N=32), evaluate boundary dispersion, and apply the temporal blend of Section 6. A final pass copies the output into the `RGBA32F` history buffer. The hero hash $h(\mathbf{x})$ of Eq. 10 uses $\text{frac}(\sin(\mathbf{x}\cdot(12.9898, 78.233))\cdot 43758.5453)$ with $g = 0.6180339$. History is additionally cleared on any camera movement (translation or rotation), preventing view-dependent smearing.

8. Results

All timings and figures are at 1920×1080 on an NVIDIA RTX 3070 Laptop GPU.

Experimental setup. The beam specimen is a $1.0\times0.2\times0.1$ box viewed by a pinhole camera at $(1.0, 0.4, 2.0)$ looking at $(0, 0.4, 0)$ with Falcor's default 21 mm focal length (59° vertical field of view at 16:9); the disk (radius 0.5, thickness 0.1) uses the same camera; the FEM plate is a unit cube viewed from $(0, 0, 2.5)$. Unless stated otherwise we use $C = 0.001$, polariser angle 45° and force magnitude 300. Dynamic-load experiments sweep the load sinusoidally along the beam for 60 frames, then hold for 50. For all error metrics, reference and test runs share a fixed path-tracer seed so the background is identical and the metric isolates the fringe layer, and SSIM/PSNR use double precision; over the 110 frames of Figure 3 our SSIM agrees with scikit-image to within 4×10^{-4} at a matched 11×11 window. Timings for Table 2 use a fixed benchmark view of the beam, $(0.24, 0.4, 0.48)$ with force 75, chosen so the deterministic baseline runs near its Table 1 cost. Released scenes and capture scripts reproduce every configuration exactly.

8.1. Performance

For analytical fields the physics pass dispatches over all voxels each frame; for pre-baked FEM data it is bypassed and the texture read directly. The physics pass is resolution-independent at 0.05 ms; the optics pass scales linearly with pixel count. At 1080p the deterministic 16λ baseline on a 128^3 grid (16 sequential scalar passes, mirroring prior work) takes 19.5 ms; our hero pass on the same grid takes 2.28 ms. The isolated pass cost measures our algorithmic contribution; global FPS also includes VBuffer ray generation, accumulation and tone mapping.

Because our static FEM ingestion matches Bußler et al.'s methodology, the 128^3 FEM row of Table 1 is a direct, apples-to-apples baseline. Replacing the deterministic 16-wavelength loop with our 4-wavelength SIMD estimator gives an **8.55× algorithmic speedup** (19.5 vs 2.28 ms) on identical hardware and grid. Against the published GTX 760 timing we demonstrate an **over 100× absolute speedup** (231 vs 2.28 ms), combining algorithmic improvement with a decade of hardware progress to break the real-time barrier for spectral photoelasticity.

Wavelength-count ablation. The speedup is not proportional to wavelength count: a 4× sample reduction alone would predict 4×, not 8.55×. The gain is architectural: packing the 4 stratified wavelengths into a single `float4` march lets the wavelength-independent per-step work (the trilinear stress fetch, the fast-axis angle and its trigonometry) be evaluated once and shared, whereas the baseline repeats it in each of 16 sequential scalar marches. Cost therefore tracks marches, not wavelengths. Table 2 isolates this on the beam scene (128^3, 32 steps, 1080p): at one march, widening the packet from 4 to 8 to 16 wavelengths leaves cost at 1.02×

Table 1: *GPU execution time for the isolated `PhotoelasticityPass` (ms), with global pipeline FPS. †Bußler et al. [BES15] (GTX 760; context only).*

Config	Grid	λ	Steps	Pass Cost	Global FPS
Ours (Analytic Fast)	16^3	4	32	0.51	240
Ours (Analytic)	32^3	4	32	0.78	200
Ours (Analytic HQ)	32^3	4	64	1.22	195
Ours (FEM Kirsch)	128^3	4	32	2.28	184
Deterministic	128^3	16	32	19.5	44
Bußler†	—	16	—	231	4.3
					VR Budget (90Hz): 11.1 ms global

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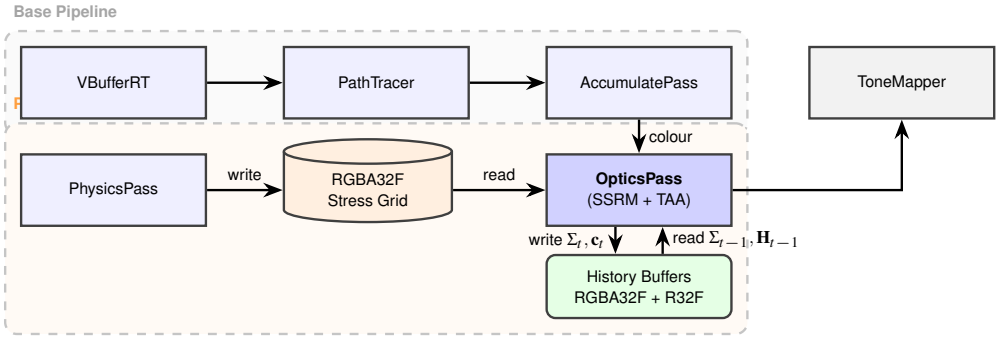


Figure 2: *Pipeline overview. The physics sub-pass either evaluates the analytical stress grid per frame or fetches pre-computed FEM data; the optics sub-pass reads it via trilinear sampling. Temporal feedback stays entirely within* `PhotoelasticityPass`.

Table 2: *Spectral-cost scaling on the beam scene (*128^3*, 32 steps, 1080p) at a fixed view, normalised to hero-4. "Marches" counts stress-volume traversals. Each entry is the median of ten interleaved rounds of 100 frames, each ratio formed within a round so boost-clock drift cancels; the ratios span* 0.99–1.10*,* 1.17–1.25*,* 3.05–3.66 *and* 12.10–13.70*, so the* 8λ *row is statistically indistinguishable from the* 4λ *row.*

Config	λ	Marches	Rel. cost
Hero quad (ours)	4	1 vector	1.00×
Hero, two quads	8	1 vector	1.02×
Hero, four quads	16	1 vector	1.20×
Hero, scalar loop	4	4 scalar	3.43×
Deterministic	16	16 scalar	13.23×

and 1.20×, while the same 4 wavelengths as sequential scalars cost 3.43× and the deterministic 16 cost 13.23×, close to proportional to march count. The most direct comparison is the 16λ hero row against the 16λ deterministic row: at an *identical* wavelength count the packed march is 10.8× cheaper (median of ten rounds, range 10.2–11.4), so the gain is not from sampling fewer wavelengths. Because absolute cost depends on screen coverage, these are within-experiment ratios at one fixed view, not the cross-scene FEM comparison behind Table 1's 8.55×.

8.2. Spectral convergence and TAA stability

We measure structural convergence of the estimator against a converged deterministic 16-wavelength reference. In Figure 3 (Left) the single-sample per-pixel estimate starts at a noisy 20.29 dB and converges smoothly to 50.86 dB under temporal accumulation.

We then validate the reprojection by tracking SSIM against a per-frame deterministic reference during a dynamic load event (Figure 3, Right), reporting the full frame and the fringe-bearing crop separately: the full frame bounds overall fidelity; the crop isolates the region where fringes actually move. With the background at a fixed seed, both scopes measure the fringe layer, not residual background noise. Our gate never falls below 0.989 full-frame or 0.940 on the crop, and neither method drops below 0.9 in either scope. Through the load sweep our gate remains above standard TAA on every statistic (crop mean 0.967 vs 0.954, minimum 0.940 vs 0.908; full-frame mean 0.994 vs 0.992). The curve shapes are as informative as their levels: standard TAA oscillates, recovering whenever the load decelerates and it briefly benefits from static accumulation, then collapsing while the load moves fastest as stale fringes smear into structurally wrong positions, the artifact that misleads visual stress interpretation. Our gate rejects that history where the stress trace changes, trading ghosting for unbiased zero-mean noise at correct fringe positions. Once the load stabilises both modes reach the same accumulation floor within a few frames; visually, however, standard TAA's fringes persist for 25 to 28 frames (Figure 4), a structural error SSIM penalises far more weakly than the noise it replaces.

8.3. TAA ablation

Figure 4 ablates the temporal strategies during an interactive load change. Disabling history ($w = 0$) gives a spectrally correct but noisy image, the raw 1-spp Monte Carlo output. Standard geometric TAA ($w = 0.95$) denoises effectively but introduces severe ghosting from stale fringe patterns. Our gate combines the strengths of both: it rejects history at the shifting stress field's boundaries, preventing ghosting, and resumes accumulation once the event concludes.

8.4. Joint geometric, view, and stress dynamics

Interactive use combines several changes at once, so we evaluate under simultaneous geometric, view and stress dynamics: the specimen rotates in-plane 0° to 30° over frames 1–90 (as on a polariscope stage, via a scene node animation), the camera orbits ∼34° over the same frames, and the sinusoidal load runs its 60-frame sweep and 50-frame hold. Rotation requires no change to the estimator or the gate: rays transform rigidly into the specimen frame before marching, and the rotation angle adds to the fast-axis angle α (Section 6). Figure 5 tracks full-frame SSIM against the per-frame deterministic reference. SSIM never falls below 0.993 and no ghosting occurs: while the pose changes, history is conservatively discarded and the image remains unbiased single-sample noise at correct fringe positions; once motion stops, accumulation resumes and error decays to the static floor. Under rigid pose change the two modes coincide exactly, since a moving camera invalidates history for both; the gate's contribution is specific to the geometrically static but optically dynamic case of Section 8.2. The curves sepa-

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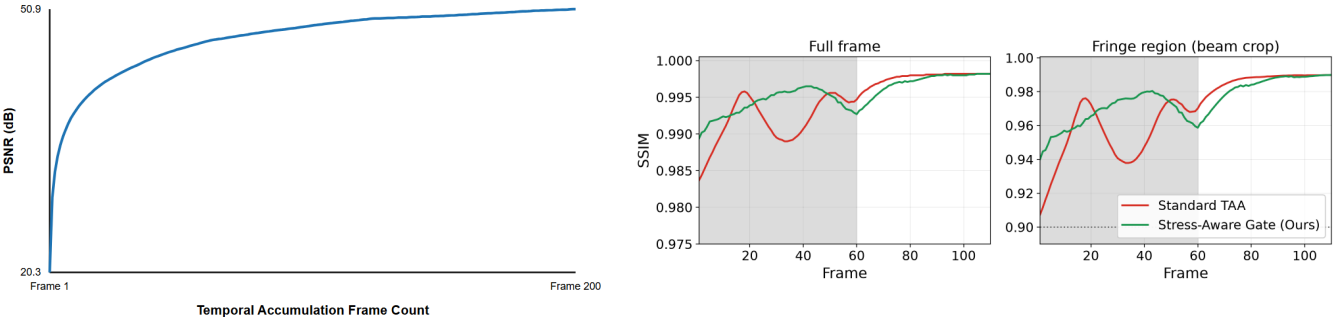


Figure 3: ***Left:*** *spectral convergence of the Hero-Wavelength estimator, 20.29 dB to 50.86 dB against the deterministic 16-wavelength reference.* ***Right:*** *SSIM during a dynamic load event (shaded, frames 1–60) and stabilisation (61–110), measured full-frame and on the fringe-region crop with double-precision SSIM and a fixed-seed background so the metric isolates the fringe layer.*



Figure 4: *TAA comparison during interactive load change.* ***Left:*** $w = 0$ *(no history): spectrally correct but noisy.* ***Centre:*** $w = 0.95$ *always: ghost of previous fringe pattern heavily smeared and visible for* ***25 to 28 frames*** *after the load moved.* ***Right:*** *stress-aware gate: history rejected at the load change, ghost-free, re-converging immediately.*

rate only when the load alone resumes (frames 111–120), and standard TAA scores marginally higher there (0.998 against 0.997): it still holds a smooth but stale ghost where our gate has already switched to unbiased noise, the trade-off of Section 8.2. Fringe patterns ride rigidly with the specimen, and the isoclinics shift with the specimen-to-polariser angle as polariscope theory requires. These analytical fields are defined on the specimen's reference geometry, so the specimen rotates rigidly without deforming; deforming FEM fields remain future work (Section 9).

8.5. Grid resolution ablation

Optimal grid resolution correlates with the field's spatial frequency. For smooth, low-frequency distributions such as the Timoshenko beam, hardware trilinear interpolation perfectly reconstructs the field between voxel centres: Figure 6 shows the 16^3 grid already achieving interior fringe-order error below 0.4% at the neutral axis, with the expected boundary quantisation at the extreme fibres (Table 3).

Even with stress singularities under point loads, as in the Flamant disk, the disparity is minimal: 64^3 is sharper at the load poles, but 16^3 resolves the vast majority of the interior identically, with only negligible blurring at the extreme stress concentrations (Figure 7). Sparse grids (16^3 to 32^3, under 1 MB VRAM) are therefore efficient and visually sufficient for interactive use, scaling to 128^3 only for rigorous analysis of high-frequency topologies.

8.6. Validation

We validate along two axes, quantitative against analytical solutions and qualitative against experimental photographs, separating simulation error from rendering artifacts. The analytical models (Timoshenko beam and Flamant disk) establish interactivity and temporal stability (Sections 8.2 and 8.4). To show that the ray-marcher handles complex topologies and high-frequency singularities, we examine the classical Kirsch problem (a plate with a central circular hole under uniaxial tension) by ingesting a static FEM tensor field, intentionally matching the baseline of Bußler et al. [BES15].

Figure 8 compares our volume against the published experimental reference (Figure 1.18a of Ramesh [Ram00]), captured in a real dark-field circular polariscope. We replicate that setup by initialising the incident ray with a circularly polarised Jones vector $\mathbf{E}_{\text{in}} = \frac{1}{\sqrt{2}}[1, i]^T$ and extracting the crossed circular intensity $I = \frac{1}{2}|E_x + iE_y|^2$, eliminating directional isoclinics to isolate pure isochromatics. Our raycaster reproduces the reference's structural features (the dense butterfly-wing concentration zones and wide sweeping arcs), confirming that the pipeline translates FEM tensors into the correct fringe pattern under real-time constraints. While the 128^3 grid reproduces the global structure, minor aliasing remains visible in the extremely dense butterfly-wing regions directly adjacent to the singularity; a finer voxel resolution (e.g. 512^3) would further resolve these extreme concentrations.

We also verify our voxelised grid interpolation against the exact Timoshenko solution. In Table 3 the rendered fringe orders deviate by less than 0.4% from analytic values at the neutral axis, rising to 6.25% at the boundaries due to hardware clamp-to-edge interpolation on the discrete 16^3 grid, in line with expected quantisation limits. For diametral compression (Figure 9) the system reproduces the classic Flamant figure-eight neutral-point pattern and dense fringe crowding at the load poles.

8.7. Polariscope sweep

Figure 10 (enlarged in the supplementary) shows the crossed polariser and analyser rotated together. As continuous photoelastic

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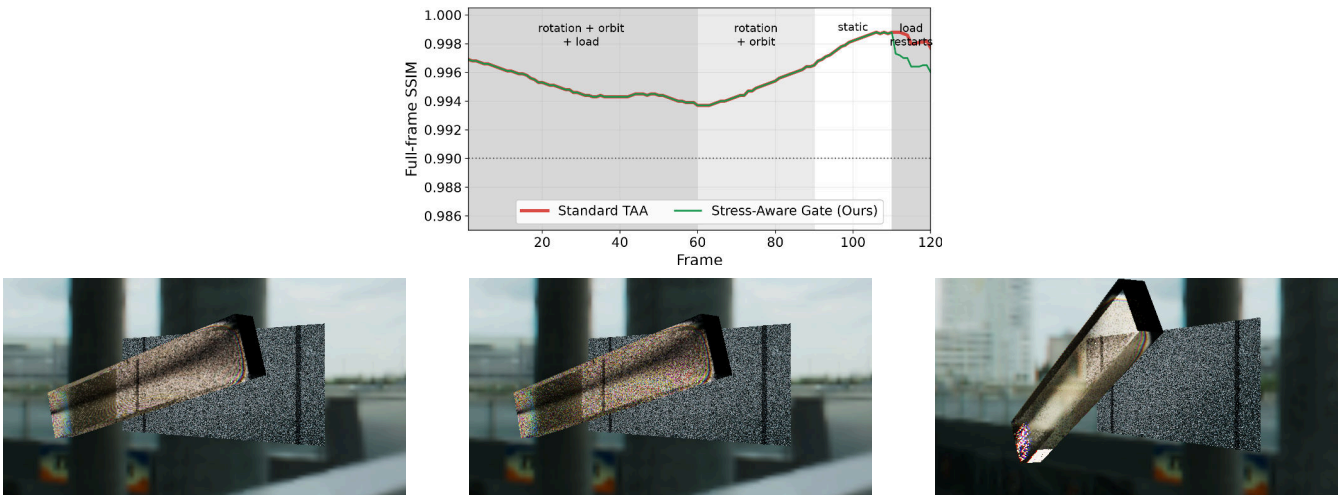


Figure 5: *Joint geometric + view + stress dynamics: specimen stage rotation (0°–30°) with a simultaneous camera orbit and dynamic sine load.* ***Top:*** *full-frame SSIM; shaded bands mark the dynamics phases.* ***Bottom, left to right:*** *deterministic reference at frame 45 (all three dynamics active); ours at frame 45, unbiased noise at correct fringe positions and no ghosting; ours at frame 110 after motion stops, re-converged. The background speckle is the fixed-seed path-traced environment shared by reference and test.*

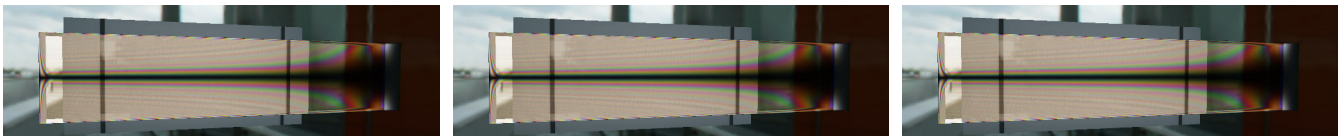


Figure 6: *Grid resolution ablation evaluated on the beam's interior volume, at* 16^3*,* 32^3 *and* 64^3 *(left to right). The three are visually indistinguishable for this nearly linear, low-frequency field.*

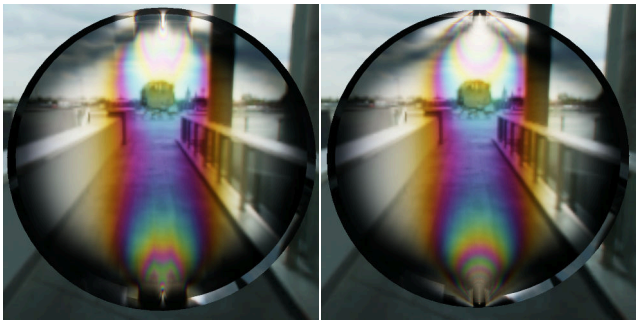


Figure 7: *Impact of grid resolution on high-frequency stress singularities:* 16^3 *(left) vs* 64^3 *(right).*

theory predicts, the isoclinic fringes (dark bands where the principal stress axes align with the polariser) rotate in tandem, sweeping 0° to 90°. This confirms the vectorised Jones integration tracks the polarisation state without artificial coordinate bias.

9. Discussion

Load oscillation. Fast continuous oscillation of the load point forces perpetual history rejection, and the viewer sees raw 1-spp noise until the load stabilises. This is acceptable for slow, deliberate load application but would hamper a physics simulation applying rapid oscillatory loads. A natural extension is to estimate fringe

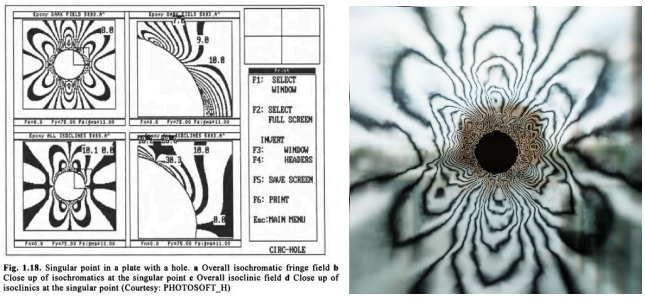


Fig. 1.18. Singular point in a plate with a hole. a Overall isochromatic fringe field b Close up of isochromatics at the singular point e Overall isoclinic field d Close up of isoclinics at the singular point (Courtesy: PHOTOSOFT_H)

Figure 8: *Physical validation of the Kirsch problem.* ***Left:*** *the experimental benchmark (Ramesh [Ram00]), a dark-field circular polariscope under a monochromatic sodium vapour lamp.* ***Right:*** *our real-time* 128^3 *FEM volume in the same simulated mode, matching the isochromatic field of the benchmark's top row.*

velocity from the stress gradient and warp the history buffer instead of discarding it. Composing motion-vector reprojection with the stress gate would likewise let history persist through continuous motion of the specimen or the camera, rather than falling back to our conservative full rejection.

Interactive finite element solvers. This paper focuses on real-time optical rendering of stress fields and the temporal aliasing that interactivity introduces. Our physics pipeline evaluates continuous analytical fields in real time, while for complex boundary prob-

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Table 3: *Fringe order at six beam positions: analytic vs. rendered (16^3 grid).*

$(x/L,\ y/H)$	N_{analytic}	N_{rendered}	Error (%)
(0.25, 0.00)	1.125	1.121	0.39%
(0.25, 0.50)	8.438	7.910	6.25%
(0.50, 0.00)	1.125	1.121	0.39%
(0.50, 0.50)	5.625	5.273	6.25%
(0.75, 0.00)	1.125	1.121	0.39%
(0.75, 0.50)	2.813	2.637	6.25%

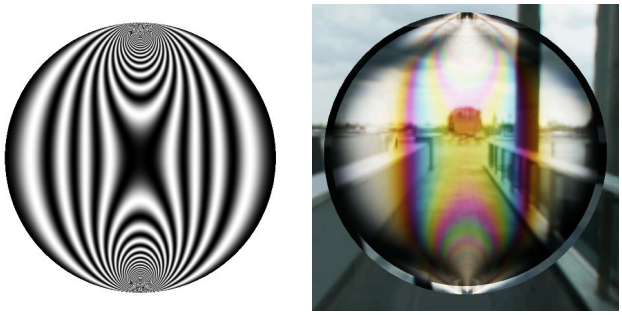


Figure 9: *Disk validation. **Left:** analytical Flamant isochromatic prediction [Fro48]. **Right:** our render. The figure-eight neutral-point pattern and fringe crowding at the load poles are successfully reproduced.*

lems (e.g. the Kirsch validation) it retrieves static pre-computed FEM tensor grids. Expanding the physics sub-pass to solve dynamic FEM simulations on the GPU would let users deform complex topologies interactively while retaining stress-aware temporal stability. This is a key area for future work.

Internal refraction. Cauchy dispersion is evaluated at the exit surface only. For the PMMA samples we use ($\Delta n \approx 0.021$), the maximum internal angular deviation along the beam axis is well under a pixel, so the straight-line march is a good approximation. Thicker objects or larger stress-optic coefficients would require non-linear ray marching.

Scattering. Jones Calculus assumes fully polarised, non-scattering light. Cloudy or translucent materials would require a Mueller-Stokes treatment, replacing the 2×2 complex Jones matrix with a 4×4 real Mueller matrix and substantially increasing per-step integration cost.

10. Conclusion

We have shown that the photoelastic spectral integral can be formulated as Monte Carlo estimation. Adapting Hero Wavelength Sampling to the Jones Calculus domain yields a real-time-capable estimator with no accuracy penalty under temporal accumulation. Our second contribution, a stress-aware history rejection gate, addresses a critical failure mode of geometric TAA that would otherwise make interactive use impractical. The result is a renderer agnostic to the physics source, visualising dynamic analytical fields and complex FEM volumes in real time. The gate also highlights

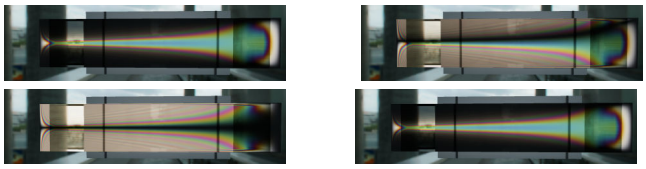


Figure 10: *Polariser sweep: 0°, 22.5°, 45°, 90°. Isoclinic fringes (dark bands at stress principal-axis directions) rotate with the analyser, as expected from polariscope theory.*

a broader open challenge: reprojecting volume effects that are geometrically static but optically dynamic, a case standard reprojection literature does not address.

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